

Fundamental Concepts

- ☐ BASIC QUANTITIES
- ☐ COORDINATE SYSTEMS



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ME101 (Spring 2017)

Fundamental Concepts
Basic Mech. Engg.
Basic Quantities
Coordinate Systems



Basic Quantities

MEASURING OBJECTS

How do we measure the size of a physical object?

Lengths & Angles

What are the units?

Meter - Radians

How long is 1 Meter?

One ten-millionth of the distance from the Earth's equator to the North Pole

Length of the path travelled by light in vacuum during a time interval of $1/299,792,458$ of a second




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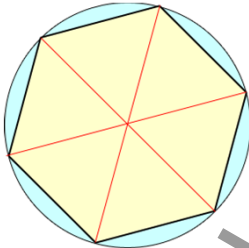


Basic Quantities

MEASURING OBJECTS

**Why a circle has 360 degrees?
Why not 248 or 623 degrees?**

Babylonians subdivided the circle using the angle of an equilateral triangle as the basic unit and further subdivided the latter into 60 parts



Which one is greater degree or radian?

1 Radian = 57.3 Degrees
Radian > Degrees

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Basic Quantities

MEASURING OBJECTS

How much is one Radian?

Numerically equal to the length of a corresponding arc of a unit circle

Length of arc (l) = Radius (r) x Angle (θ)



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Basic Quantities

RADIUS OF EARTH

What is the mean radius of earth?

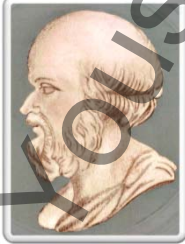
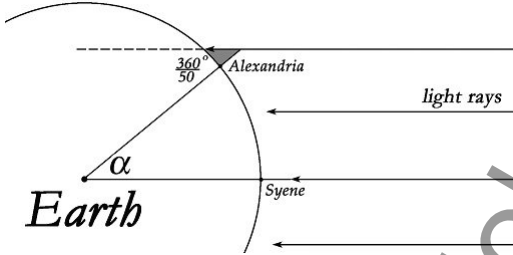
6371 Km

Who calculated it?

Eratosthenes

How?

$l = r \theta$



Eratosthenes
(276BC - 194BC)
Greek mathematician,
geographer, poet,
astronomer



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Basic Quantities

MEASUREMENT QUANTITY

How do we measure that amount of matter in an object?

Mass

Units?

Kilogram (Kg)

How is it defined?

How much is 1Kg?

1 gram is the mass of 1 cubic centimeter of water

How is it measured?

Inertial Balance

What is the mass of earth?

$5.972 \times 10^{24} \text{ Kg}$

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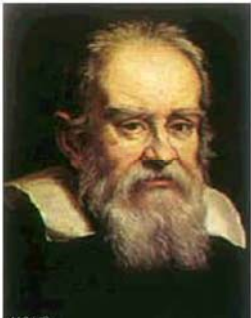


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MASS OF EARTH

How was mass of the Earth measured?

$$g = \frac{GM_e}{R^2}$$



Galileo Galilei
(1564-1642)

$$g = 9.81 \text{ m/s}$$



Isaac Newton
(1642-1726)

$$F = \frac{Gm_1m_2}{r^2}$$



Henry Cavendish
(1731-1810)

$$G = 66.73 \times 10^{-12} \frac{\text{m}^3}{\text{Kg} \cdot \text{s}^2}$$

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Basic Quantities

MEASURING PUSH/PULL

Can we measure "Push" or "Pull"?



PUSH



PULL

Yes! Force in Newton

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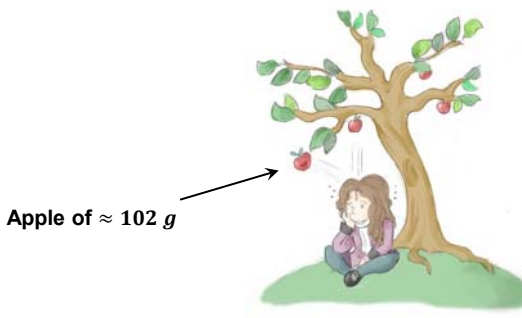
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Basic Quantities

MEASURING PUSH/PULL

How much is the push/pull of one Newton force?



Apple of $\approx 102\text{ g}$

What is the pulling effect of Earth on an object called?

Weight

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Basic Quantities

MEASURING PUSH/PULL

Is weight of an object constant?

No!

Why?

$$F = \frac{Gm_1m_2}{r^2}$$

G = Universal Constant of Gravitation = 66.73×10^{-12}

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Coordinate Systems

INTRODUCTION

What do we need to specify the position of an object in 3D space ?

A reference
A convention for representation

What do we get by combining the above mentioned entities?

A coordinate system

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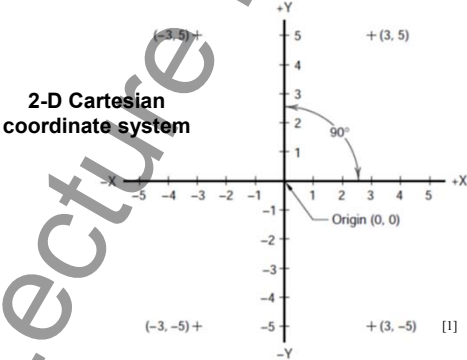


Coordinate Systems

HISTORY

for every point in space, a set of real numbers can be assigned, and for each set of real numbers, there is a unique point in space

2-D Cartesian coordinate system





Renè Descartes (1596–1650)
French Mathematician
“Cartesius”

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 307.

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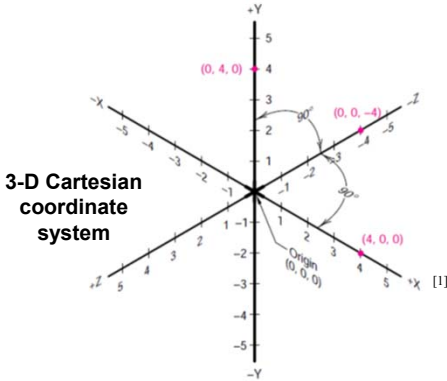
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Coordinate Systems

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3-D Cartesian coordinate system



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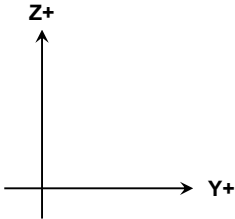
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Coordinate Systems

CARTESIAN COORDINATE SYSTEM



In which direction is X+ axis?
In the plane or out of the plane?
Out of the plane

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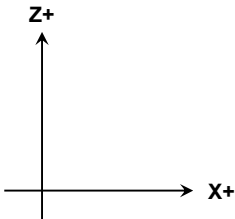
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Coordinate Systems

CARTESIAN COORDINATE SYSTEM



In which direction is Y+ axis?
In the plane or out of the plane?

In the plane

How do you know for sure?
What is the rule that you are applying?

Right Hand Rule

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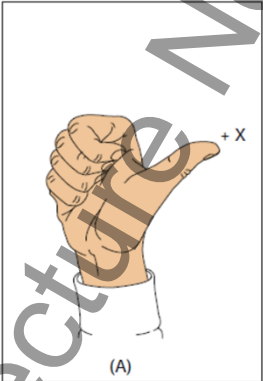
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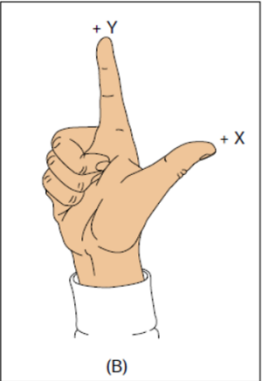


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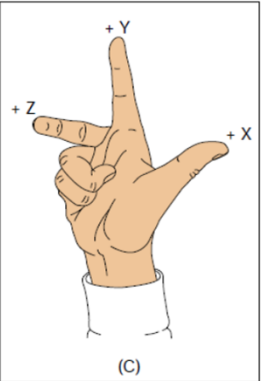
RIGHT HAND RULE



(A)



(B)



(C)

[1]

Right-Hand Rule

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 310.

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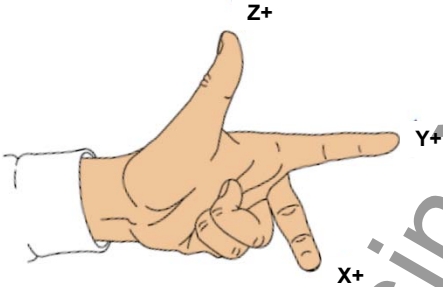
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RIGHT HAND RULE



Left Hand Rule (Not Documented)

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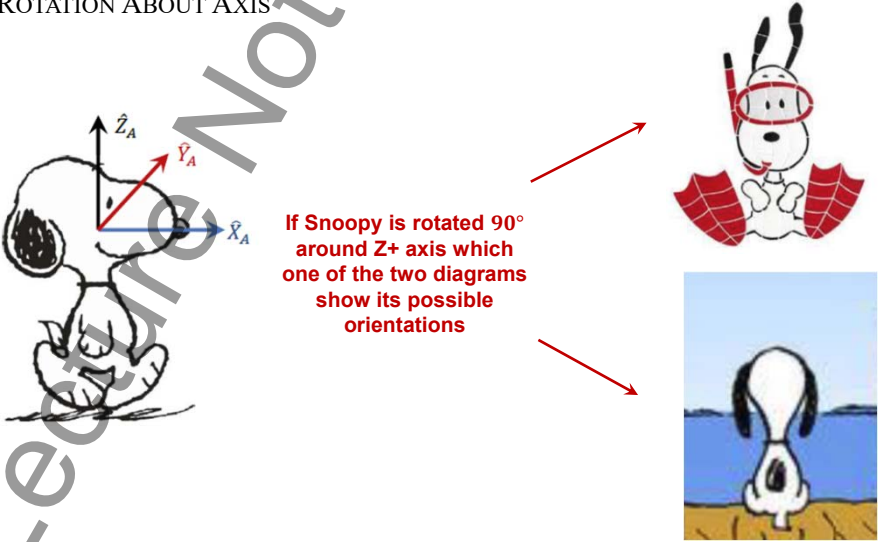
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Coordinate Systems

ROTATION ABOUT AXIS



If Snoopy is rotated 90° around Z+ axis which one of the two diagrams show its possible orientations

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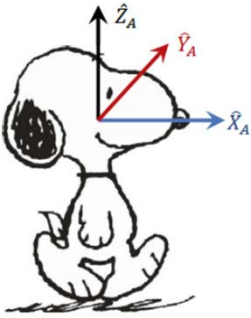
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ROTATION ABOUT AXIS



If Snoopy is rotated 90° around Y+ axis which one of the two diagrams show its possible orientations



How do you know for sure?
What is the rule that you are applying?

Right Hand Rule

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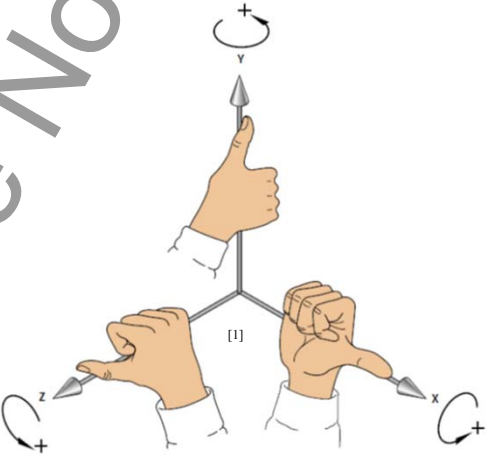
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ROTATION ABOUT AXIS – RIGHT HAND RULE



Positive directions of rotation on each axis.

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 310.

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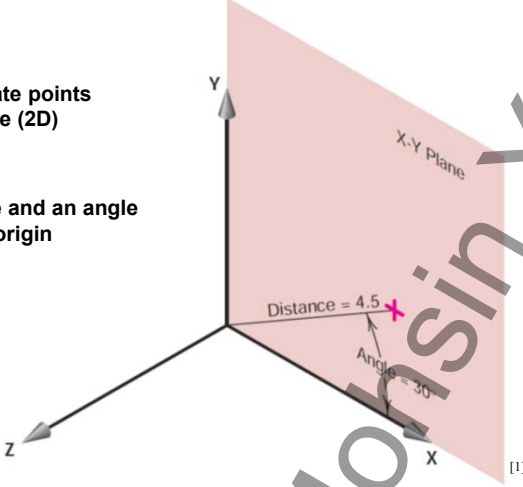


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POLAR COORDINATES

Is used to locate points in any plane (2D)

Specify a distance and an angle from the origin



[1]

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 311.

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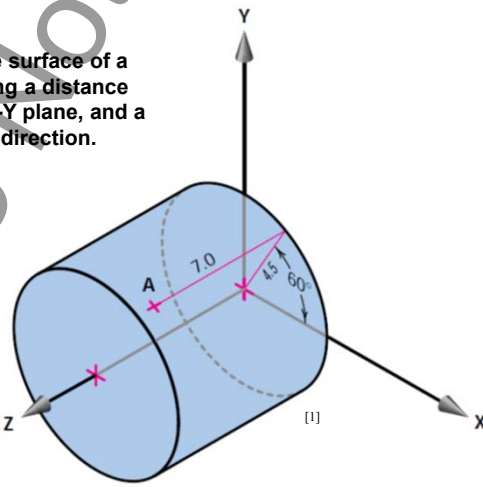
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CYLINDRICAL COORDINATES

locate a point on the surface of a cylinder by specifying a distance and an angle in the X-Y plane, and a distance in the Z direction.



[1]

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 311.

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SPHERICAL COORDINATES

locate a point on the surface of a sphere by specifying an angle in one plane, an angle in another plane, and one length.

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 312.

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Take-Home Message

CONCEPT

Right Hand Rule

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